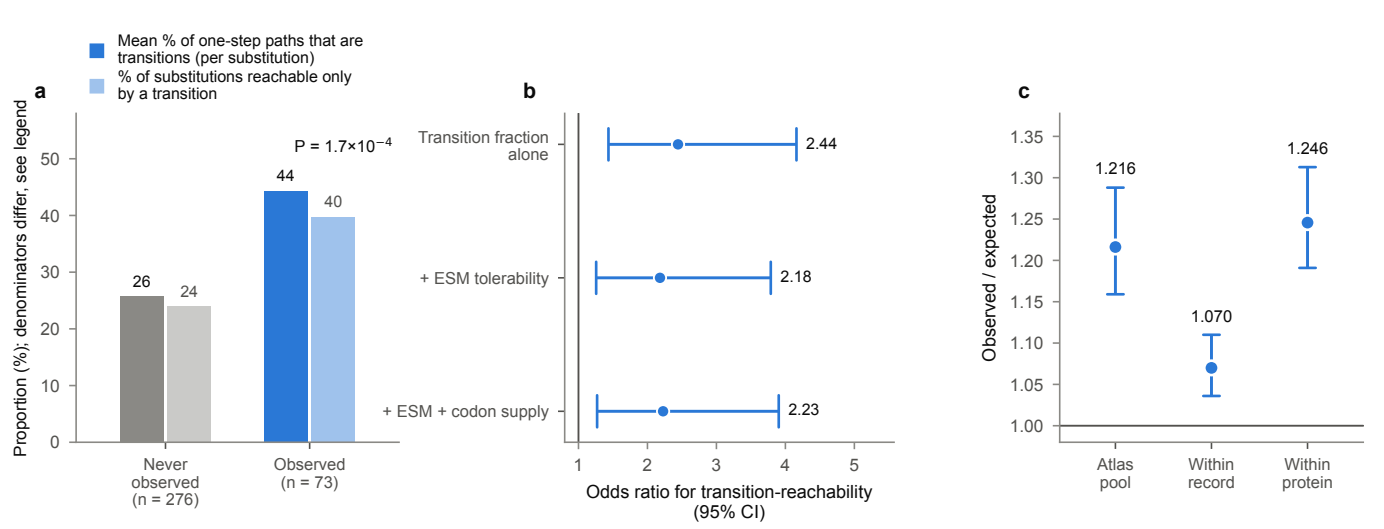
a
Mean % of one-step paths that are transitions (per substitution)
% of substitutions reachable only by a transition
Proportion (%); denominators differ, see legend
P = 1.7×10−4
26
24
44
40
Never observed (n = 276)
Observed (n = 73)
b
Transition fraction alone
+ ESM tolerability
+ ESM + codon supply
2.44
2.18
2.23
Odds ratio for transition-reachability (95% CI)
c
Observed / expected
1.216
1.070
1.246
Atlas pool
Within record
Within protein